

$$(1) \quad \ddot{x} + 5\dot{x} + 6x = t^3 e^{-t} + \cos(3t)$$

$$s^2 X(s) + 5sX(s) + 6X(s) = \frac{3!}{(s+1)^4} + \frac{s}{s^2+9}$$

$$-sX(0) - \dot{X}(0) - X(0)$$

$$X(s) (s^2 + 5s + 6) - \frac{s}{3} + 1 - \frac{s}{3} = X(s) (s^2 + 5s + 6) - \frac{s}{3} - \frac{2}{3}$$

$$= \frac{6}{(s+1)^4} + \frac{2}{s^2+9}$$

$$X(s) (s+2)(s+3) - \frac{(s+2)}{3} = \frac{6}{(s+1)^4} + \frac{2}{s^2+9}$$

$$\Rightarrow X(s) = \frac{1}{3(s+3)} + \frac{6}{(1+s)^4 (s+2)(s+3)} + \frac{s}{(s^2+9)(s+2)(s+3)}$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\frac{1}{3} e^{-3t} \quad a \quad b$$

$$a = \frac{6}{(1+s)^4 (s+2)(s+3)}$$

$$= \frac{A}{1+s} + \frac{B}{(1+s)^2} + \frac{C}{(1+s)^3} + \frac{D}{(1+s)^4} + \frac{E}{(s+2)} + \frac{F}{(s+3)}$$

$$\Rightarrow \frac{6}{(s+2)(s+3)} = A(1+s)^3 + B(1+s)^2 + C(1+s) + D + \frac{E(1+s)^4}{s+2} + \frac{F(1+s)^4}{s+3}$$

$$s = -1 \Rightarrow D = \frac{6}{2} = 3$$

$$s = -2 \Rightarrow E = \frac{6}{1} = 6$$

$$s = -3 \Rightarrow F = \frac{6}{-1} = -6$$

$$L = \frac{6}{(s+2)(s+3)} = \frac{6}{s+2} - \frac{6}{s+3}$$

$$L' = \frac{-6}{(s+2)^2} + \frac{6}{(s+3)^2}$$

$$s = -1 \Rightarrow C = \frac{-6}{1} + \frac{6}{4}$$

$$= \frac{3}{2} - 6 = -\frac{9}{2}$$

$$L'' = \frac{+12}{(s+2)^3} - \frac{12}{(s+3)^3}$$

$$s = -1 \Rightarrow 2B = \frac{12}{1} - \frac{12}{8}$$

$$= 12 - \frac{3}{2} = \frac{21}{2} \quad B = \frac{21}{4}$$

$$L''' = \frac{-36}{(s+2)^4} + \frac{36}{(s+3)^4}$$

$$s = -1 \Rightarrow 6A = \frac{-36}{1} + \frac{36}{16} = \frac{-36 \times 15}{16}$$

$$\Rightarrow A = -\frac{45}{8}$$

$$\Rightarrow L^{-1}(a) = -\frac{45}{8}e^{-t} + \frac{21}{4}te^{-t} - \frac{9}{4}t^2e^{-t} + \frac{t^3}{2}e^{-t} + 6e^{-2t}$$

$$b = \frac{s}{(s^2+9)(s+2)(s+3)} = \frac{A}{s+2} + \frac{B}{s+3} + \frac{Cs+D}{s^2+9}$$

$$s = -2 \Rightarrow \frac{-2}{(9+4)(1)} = A \frac{1}{(9+4)} \Rightarrow A = \frac{-2}{13}$$

$$s = -3 \Rightarrow \frac{-3}{(9+9)(-1)} = B \Rightarrow B = \frac{1}{6}$$

$$s = 0 \Rightarrow 0 = \frac{-2}{13} \times 3 \times 9 + \frac{1}{6} \times 2 \times 9 + D \times 6$$

$$\Rightarrow 6D = \frac{54}{13} - 3 = \frac{15}{13} \Rightarrow D = \frac{5}{26}$$

$$s = -1 \Rightarrow -1 = \frac{-2}{13} \times 2 \times 10 + \frac{1}{6} \times 1 \times 10 - C \times 2 + \frac{5}{26} \times 2$$

$$\Rightarrow C = \frac{-1}{78}$$

$$\Rightarrow L^{-1}(b) = \frac{-2}{13}e^{-2t} + \frac{e^{-3t}}{6} - \frac{\cos(3t)}{78} + \frac{5 \sin(3t)}{78}$$

$$\Rightarrow X(t) = \frac{e^{-3t}}{3} - \frac{45}{8}e^{-t} + \frac{21}{4}te^{-t} - \frac{9}{4}t^2e^{-t} + \frac{t^3}{2}e^{-t} + 6e^{-2t} - \frac{3}{8}e^{-t}$$

$$- \frac{2}{13}e^{-2t} + \frac{e^{-3t}}{6} - \frac{\cos(3t)}{78} + \frac{5 \sin(3t)}{78}$$

$$\Rightarrow X(t) = e^{-3t} \left(\frac{1}{8} \right) + \frac{76}{13}e^{-2t} + \left(\frac{t^3}{2} - \frac{9}{4}t^2 + \frac{21}{4}t - \frac{45}{8} \right) e^{-t}$$

$$- \frac{\cos(3t)}{78} + \frac{5 \sin(3t)}{78}$$

2. Final Value Theorem

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

(a) $F(s) = \frac{10}{s(s+1)} \quad \lim_{s \rightarrow 0} sF(s) = 10$

$$F(s) = \frac{10}{s(s+1)} = \frac{((s+1) - (s))10}{(s+1)(s)} = \frac{10}{s} - \frac{10}{s+1}$$

$$\mathcal{L}^{-1}(F(s)) = f(t) = 10e^0 - 10e^{-t}$$

$$\lim_{t \rightarrow \infty} f(t) = 10$$

(b) $F(s) = \frac{(s-2)}{s(s+1)(s+2)} \quad \lim_{s \rightarrow 0} sF(s) = -1$

$$F(s) = \frac{s+1-1}{s(s+1)(s+2)} = \frac{s-1}{s(s+1)(s+2)} - \frac{1}{s(s+1)(s+2)}$$

$$= \frac{(s+1)+(s)-(s+2)}{s(s+1)(s+2)} - \frac{(s+2)-(s+1)}{s(s+1)(s+2)}$$

$$= \frac{1}{s(s+2)} + \frac{1}{(s+2)(s+1)} - \frac{1}{s(s+1)} - \frac{1}{s(s+1)} + \frac{1}{s(s+2)}$$

$$= \frac{2}{s(s+2)} - \frac{2}{s(s+1)} + \frac{1}{(s+1)(s+2)}$$

$$= \frac{s+2-s}{s(s+2)} - 2 \frac{((s+1)-(s))}{s(s+1)} + \frac{(s+2)-(s+1)}{(s+1)(s+2)}$$

$$= \frac{1}{s} - \frac{1}{s+2} - \frac{2}{s} + \frac{2}{s+1} + \frac{1}{s+1} - \frac{1}{s+2}$$

$$= -\frac{1}{s} + \frac{3}{s+1} - \frac{2}{s+2}$$

$$f(t) = -1 + 3e^{-t} - 2e^{-2t}$$

$$\lim_{t \rightarrow \infty} f(t) = -1$$

$$(C) \quad F(s) = \frac{s+2}{s(s-1)(s-2)} = \frac{A}{s} + \frac{B}{s-1} + \frac{C}{s-2}$$

$$s=0 \quad 2 = A(-1)(-2) \quad \Rightarrow \quad A=1$$

$$s=1 \quad 3 = B(1)(-1) \quad \Rightarrow \quad B=-3$$

$$s=2 \quad 4 = C(2)(2) \quad \Rightarrow \quad C=2$$

$$F(s) = \frac{1}{s} + \frac{-3}{s-1} + \frac{2}{s-2} \quad f(t) = 1 - 3e^t + 2e^{2t}$$

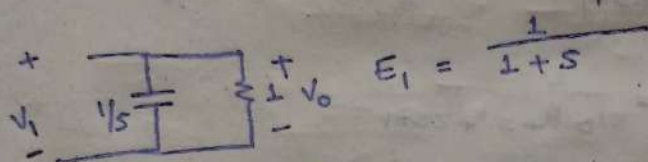
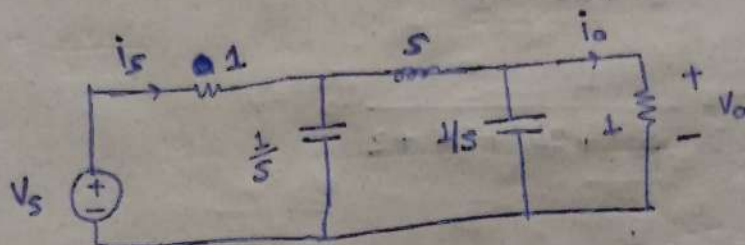
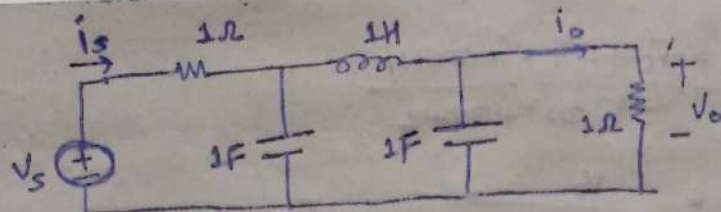
$$\lim_{s \rightarrow 0} s F(s) = \frac{2}{2} = 1$$

$$\lim_{t \rightarrow \infty} f(t) = \infty$$

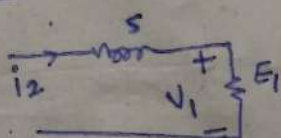
$$\text{So } \lim_{s \rightarrow 0} s F(s) \neq \lim_{t \rightarrow \infty} f(t)$$

Because the poles are in the right half plane and so the system doesn't follow the final value theorem.

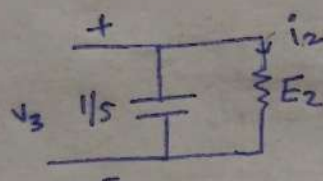
③



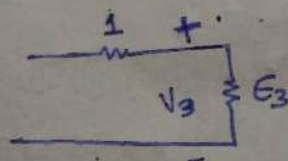
$$E_1 = \frac{1}{1+s}$$



$$E_2 = \frac{\frac{1}{s} + s + 1}{\frac{1}{s} + s + 1} = \frac{s^2 + s + 1}{1 + s}$$



$$E_3 = \frac{1}{s + \frac{1}{E_2}} = \frac{1}{s + \frac{1+s}{s^2+s+1}} = \frac{s^2+s+1}{s^3+s^2+2s+1}$$



$$E_4 = 1 + E_3 = \frac{s^3 + 2s^2 + 3s + 2}{s^3 + s^2 + 2s + 1}$$



$$i_s = \frac{V_s}{E_4} = \frac{V_s(s^3 + s^2 + 2s + 1)}{s^3 + 2s^2 + 3s + 2}$$

$$V_2 = V_s - i_s(1) = V_s - i_s$$

$$= V_s \left(\frac{s^2 + s + 1}{s^3 + 2s^2 + 3s + 2} \right) = V_3$$

$$i_2 = \frac{V_3}{E_2} = V_s \frac{(s+1)}{s^3 + 2s^2 + 3s + 2} = i_2$$

$$V_1 = V_3 - i_2 s = \frac{1 \times V_s}{s^3 + 2s^2 + 3s + 2}$$

$$V_o = V_1 \quad i_o = \frac{V_o}{1} = V_o$$

$$V_s(s) \quad I_s(s) = \frac{V_s(s^3 + s^2 + 2s + 1)}{s^3 + 2s^2 + 3s + 2}$$

$$I_o = V_o = \frac{V_s}{s^3 + 2s^2 + 3s + 2}$$

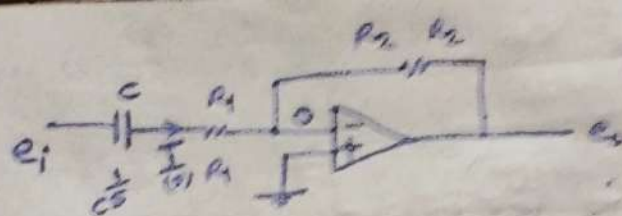
$$(a) \quad H_1(s) = \frac{V_o(s)}{V_s(s)} = \frac{1}{s^3 + 2s^2 + 3s + 2}$$

$$(b) \quad H_2(s) = \frac{V_o(s)}{I_s(s)} = \frac{1}{s^3 + s^2 + 2s + 1}$$

$$(c) \quad H_3(s) = \frac{I_o(s)}{I_s(s)} = \frac{V_o(s)}{I_s(s)} = \frac{1}{s^3 + s^2 + 2s + 1}$$

$$(d) \quad H_4(s) = \frac{I_o(s)}{V_s(s)} = \frac{1}{s^3 + s^2 + 2s + 1}$$

(4)



(a)

$$V_+ = V_- = 0$$

$$E_i(s) - \frac{I(s)}{sC} - I(s)R_1 = 0$$

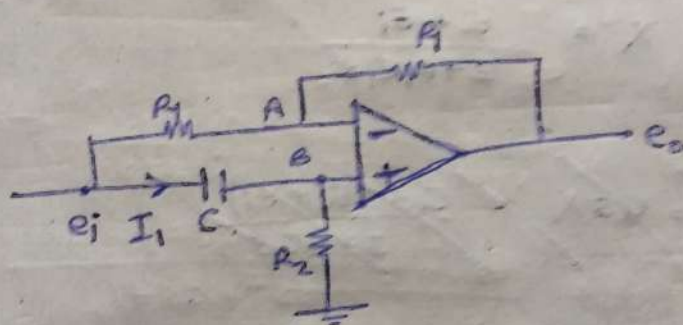
$$\Rightarrow E_i(s) = I(s) \left(\frac{1}{sC} + R_1 \right)$$

$$E_o = 0 - I(s)R_2$$

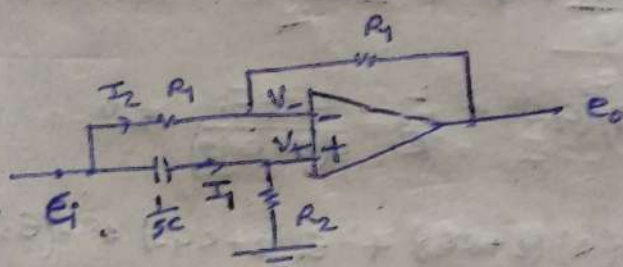
$$= \frac{-E_i(s)R_2}{\frac{1}{sC} + R_1} = E_o$$

$$\frac{1}{sC} + R_1$$

$$\Rightarrow \frac{E_o}{E_i} = \frac{-R_2}{R_1 + \frac{1}{sC}}$$



(b)



$$E_i - I_1 \left(\frac{1}{sC} + R_2 \right) = 0 \quad V_+ = I_1 R_2 = \frac{E_i R_2}{\frac{1}{sC} + R_2}$$

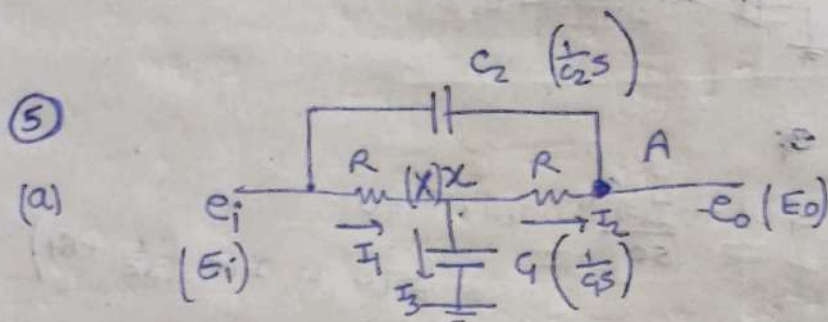
$$V_+ = V_- = \frac{E_i R_2}{\frac{1}{sC} + R_2}$$

$$I_2 = \frac{-V_- + E_i}{R_1} = \frac{E_i \left(\frac{R_2}{\frac{1}{sC} + R_2} - 1 \right)}{R_1}$$

$$E_o = V_- - I_2 R_1$$

$$= \frac{E_i R_2}{\frac{1}{sC} + R_2} + \frac{E_i R_2}{\frac{1}{sC} + R_2} - E_i = E_i \left(\frac{2R_2}{\frac{1}{sC} + R_2} - 1 \right)$$

$$\Rightarrow \frac{E_o}{E_i} = \frac{R_2 \cdot \frac{1}{sC}}{R_2 + \frac{1}{sC}}$$



$$\times \frac{X - E_i}{R} + \frac{X - E_o}{R} + \frac{(X - 0)C_1 s}{1} = 0$$

$$X \left(\frac{1}{R} + \frac{1}{R} + C_1 s \right) = \frac{E_o + E_i}{R}$$

$$X (2 + R C_1 s) = E_o + E_i$$

$$X = \frac{E_o + E_i}{2 + R C_1 s}$$

$$I_3 = \frac{X}{\frac{1}{C_1 s}} = C_1 s X = \frac{C_1 s (E_o + E_i)}{2 + R C_1 s}$$

at A

$$(E_o - E_i) C_2 s + \frac{E_o - X}{R} = 0$$

$$(E_o - E_i) C_2 s + \frac{E_o - \frac{E_o + E_i}{2 + R C_1 s}}{R} = 0$$

$$(E_o - E_i) C_2 s (R) (2 + R C_1 s) + E_o (2 + R C_1 s) - (E_o + E_i) = 0$$

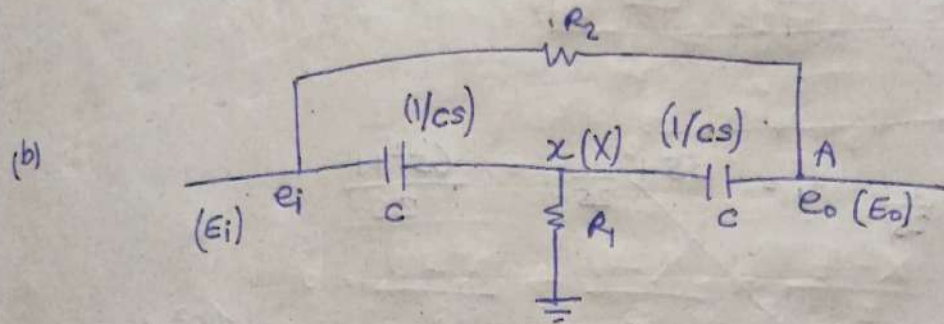
$$(E_o - E_i) (R C_2 (2 + R C_1 s)) + E_o (2 + R C_1 s) - E_o - E_i = 0$$

$$E_o (2 + R C_1 s) (R C_2 + 1) - E_i (R C_2 (2 + R C_1 s)) = 0$$

$$\frac{E_o}{E_i} = \frac{R C_2}{1 + R C_2}$$

$$E_o(RS G_2(2+RGS) + (2+RGS) + 1) = E_i(G_2SR(2+RGS) + 1)$$

$$\Rightarrow \frac{E_o}{E_i} = \frac{s^2 G_2 R^2 + 2SG_2R + 1}{s^2 G_2 R^2 + s(2G_2R + GR) + 1}$$



$$(X - E_i)CS + (X - E_o)CS + \frac{X}{R_1} = 0$$

$$\Rightarrow X = \frac{(E_o + E_i)CS}{2CS + \frac{1}{R_1}}$$

at A,

$$\frac{E_o - E_i}{R_2} + (E_o - X)CS = 0$$

$$E_o - E_i + E_o R_2 CS - X(R_2 CS) = 0$$

$$E_o(1 + R_2 CS) - E_i - \frac{(E_o + E_i)CS}{2CS + \frac{1}{R_1}}(R_2 CS) = 0$$

$$E_o(1 + R_2 CS) \left(2CS + \frac{1}{R_1}\right) - E_i \left(2CS + \frac{1}{R_1}\right) - (E_o + E_i) \frac{R_2 CS^2}{1} = 0$$

$$E_o \left((1 + R_2 CS) \left(2CS + \frac{1}{R_1}\right) - R_2 CS^2 \right) = E_i \left(2CS + \frac{1}{R_1} + R_2 CS^2 \right)$$

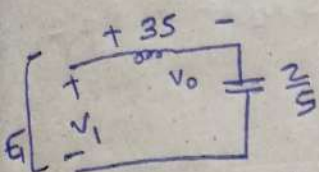
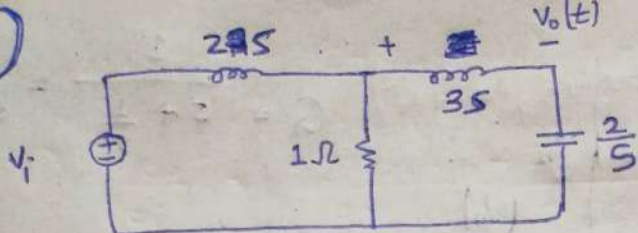
$$\frac{E_o}{E_i} = \frac{2CSR_1 + 1 + C^2 S^2 R_1 R_2}{(R_1 + R_1 R_2 CS)(2R_1 CS + 1) - C^2 S^2 R_2}$$

$$= \frac{C^2 S^2 R_1 R_2 + 2CSR_1 + 1}{2R_1^2 R_2 C^2 S^2 - C^2 S^2 R_2 + R_1 R_2 CS + 2R_1^2 CS + R_1}$$

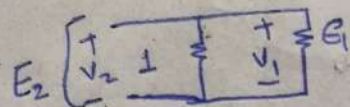
$$1 + 2CS = \frac{C^2 S^2 R_1 R_2 + 2CSR_1 + 1}{C^2 S^2 (R_1^2 R_2 - R_2) + CS(2R_1^2 + R_1 R_2) + R_1}$$

6

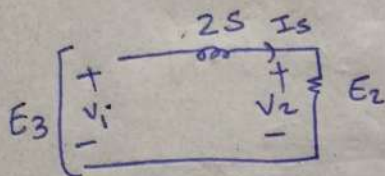
(a)



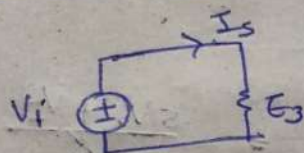
$$E_1 = 3s + \frac{2}{s} = \frac{3s^2 + 2}{s}$$



$$E_2 = \frac{1}{1 + \frac{1}{E_1}} = \frac{E_1}{1 + E_1} = \frac{\frac{3s^2 + 2}{s}}{\frac{3s^2 + s + 2}{s}} = \frac{3s^2 + 2}{3s^2 + s + 2}$$



$$E_3 = 2s + E_2 = \frac{6s^3 + 5s^2 + 4s + 2}{3s^2 + s + 2}$$



$$I_s = \frac{V_i}{E_3}$$

$$V_2 = V_i - 2s I_s = V_i \left(1 - \frac{2s}{E_3} \right)$$

$$V_1 = V_2$$

$$V_o = \frac{2/s V_1}{2/s + 3s} = \frac{2}{2 + 3s^2} V_1 = \frac{V_i \left(1 - \frac{2s}{E_3} \right) 2}{1 + 6s^2 (2 + 3s^2)}$$

$$\frac{V_o}{V_i} = \frac{2(E_3 - 2s)}{E_3 (2 + 3s^2)} = \frac{2E_2}{E_3 (2 + 3s^2)} = \frac{2(3s^2 + 2)}{3s^2 + s + 2} \cdot \frac{6s^3 + 5s^2 + 4s + 2}{3s^2 + s + 2}$$

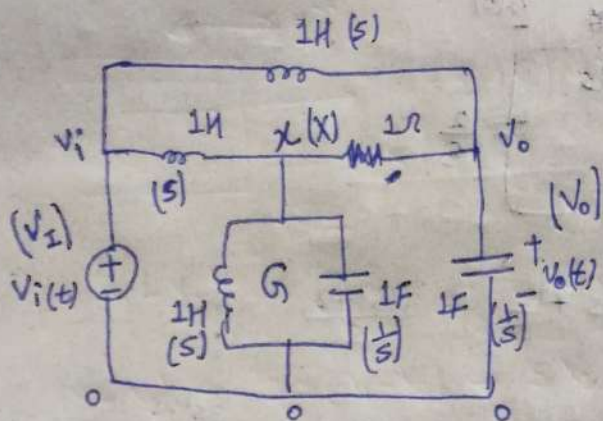
$$= \frac{3s^2}{6s^3 + 5s^2 + 4s + 2}$$

$$0 = \frac{3s^2}{6s^3 + 5s^2 + 4s + 2} - \frac{(1 + 2s + 2s^2)}{(1 + 2s + 2s^2)} = \frac{3s^2 - (1 + 2s + 2s^2)}{6s^3 + 5s^2 + 4s + 2}$$

$$0 = \left(\frac{3s^2}{6s^3 + 5s^2 + 4s + 2} - \frac{1 + 2s + 2s^2}{6s^3 + 5s^2 + 4s + 2} \right) \cdot \frac{6s^3 + 5s^2 + 4s + 2}{6s^3 + 5s^2 + 4s + 2} = \frac{3s^2 - (1 + 2s + 2s^2)}{6s^3 + 5s^2 + 4s + 2}$$

$$\left(\frac{3s^2}{6s^3 + 5s^2 + 4s + 2} - \frac{1 + 2s + 2s^2}{6s^3 + 5s^2 + 4s + 2} \right) \cdot \frac{6s^3 + 5s^2 + 4s + 2}{6s^3 + 5s^2 + 4s + 2} = \frac{3s^2 - (1 + 2s + 2s^2)}{6s^3 + 5s^2 + 4s + 2}$$

b)



$$G = \frac{1}{s + \frac{1}{s}} = \frac{s}{s^2 + 1}$$

$$\frac{X - V_I}{s} + \frac{X - V_O}{1} + \frac{X - 0}{G} = 0$$

$$\frac{X - V_I}{s} + X - V_O + \frac{Xs}{s^2 + 1} = 0$$

$$X \left(1 + \frac{1}{s} + \frac{s}{s^2 + 1} \right) = V_O + \frac{V_I}{s}$$

$$X \cdot \left(\frac{s(s^2 + 1) + s^2 + 1 + s}{(s^2 + 1)s} \right) = \frac{sV_O + V_I}{s}$$

$$X = \frac{(sV_O + V_I)(s^2 + 1)}{(s^3 + s^2 + 2s + 1)}$$

$$\frac{V_O - V_I}{s} + \frac{V_O - X}{1} + \frac{V_O - 0}{1/s} = 0$$

$$\frac{V_O - V_I}{s} + V_O - X + sV_O = 0$$

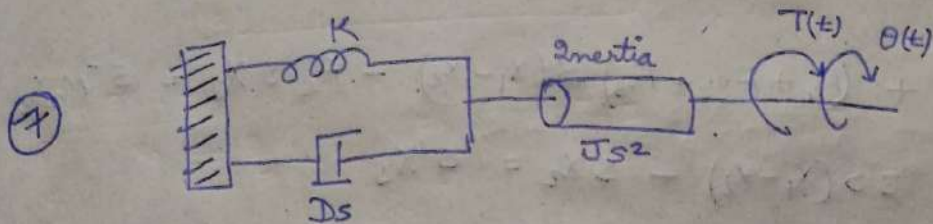
$$\frac{V_O}{s} + sV_O + V_O - X - \frac{V_I}{s} = 0$$

$$\frac{V_O}{s} + sV_O + V_O - \frac{(sV_O + V_I)(s^2 + 1)}{(s^3 + s^2 + 2s + 1)} - \frac{V_I}{s} = 0$$

$$V_O \left(\frac{1}{s} + s + 1 - \frac{s(s^2 + 1)}{s^3 + s^2 + 2s + 1} \right) - V_I \left(\frac{1}{s} + \frac{s^2 + 1}{s^3 + s^2 + 2s + 1} \right) = 0$$

$$V_O \left((s^3 + s^2 + 2s + 1) + s^2(s^3 + s^2 + 2s + 1) + s(s^3 + s^2 + 2s + 1) - s^2(s^2 + 1) \right) = V_I (s^3 + s^2 + 2s + 1 + s(s^2 + 1))$$

$$\Rightarrow \frac{V_o}{V_i} = \frac{s^2 + 2s + 2}{s^4 + 2s^3 + 3s^2 + 3s + 2}$$



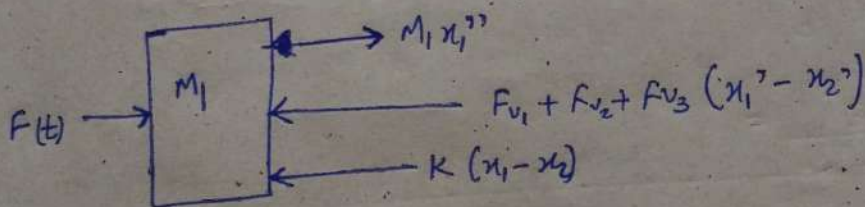
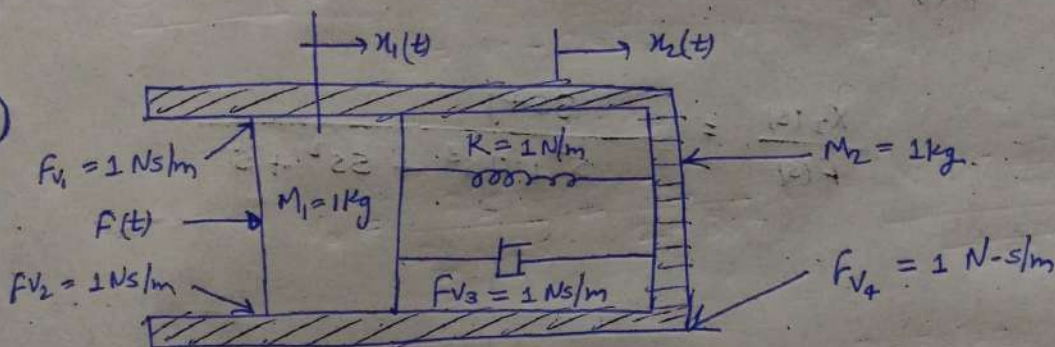
$$T(t) = T_{\text{spring}} + T_{\text{mass with inertia}} + T_{\text{viscous damping}}$$

$$= K \cdot \theta(t) + J \times \frac{d^2 \theta(t)}{dt^2} + D \times \frac{d\theta(t)}{dt}$$

$$T(s) = K\theta(s) + Js^2\theta(s) + Ds\theta(s)$$

$$= \theta(s) (K + Js^2 + Ds)$$

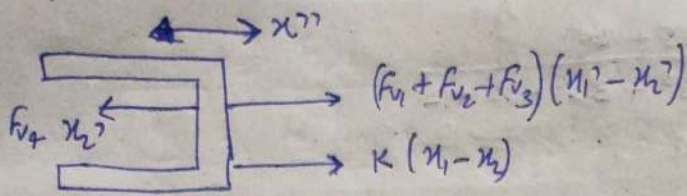
$$\frac{\theta(s)}{T(s)} = \frac{1}{Js^2 + Ds + K}$$



$$F(s) = K(x_1 - x_2) + s(F_{v1} + F_{v2} + F_{v3})(x_1 - x_2) + M_1 s^2 x_1$$

$$\Rightarrow F(s) = x_1 - x_2 + s(3)(x_1 - x_2) + s^2 x_1$$

$$F(s) = x_1(s^2 + 3s + 1) - x_2(3s + 1) \quad \text{--- (1)}$$



$$K(x_1 - x_2) + (F_1 + F_{v2} + F_{v3})s(x_1 - x_2) - F_{v4}s x_2 = s^2 M x_1$$

$$(x_1 - x_2) + 3s(x_1 - x_2) - s x_2 = s^2 x_2$$

$$\Rightarrow X_1(s) = X_2(s) \frac{(s^2 + 4s + 1)}{(1 + 3s)} \quad \text{--- (2)}$$

① and ② $\Rightarrow$

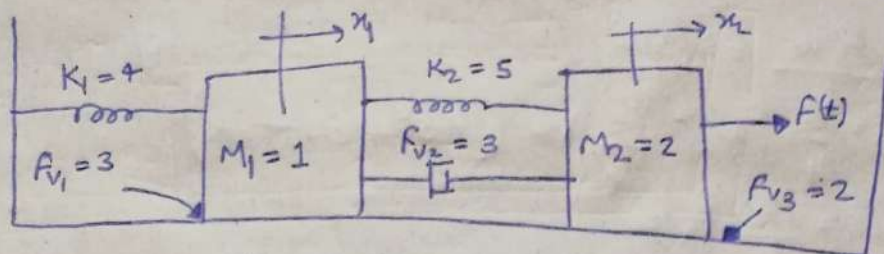
$$F(s) = X_2(s) \left(\frac{s^2 + 4s + 1}{1 + 3s} \right) (s^2 + 3s + 1) - X_2(s) (3s + 1)$$

$$\frac{X_2(s)}{F(s)} = \frac{3s + 1}{s^4 + 7s^3 + 14s^2 + 7s + 1 - (3s + 1)^2}$$

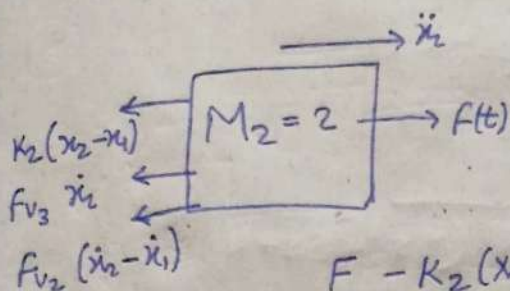
$$\frac{X_2(s)}{F(s)} = \frac{3s + 1}{s^4 + 7s^3 + 5s^2 + s}$$

$$(s^4 - 4s^3 + 4s^2 - 4s + 1) \frac{X_2(s)}{F(s)} + (s^2 - 2s + 1) \frac{X_2(s)}{F(s)} + (s^2 - 2s + 1) \frac{X_2(s)}{F(s)} + (s^2 - 2s + 1) \frac{X_2(s)}{F(s)} = 1$$

(10)



$$\frac{X_1(s)}{F(s)}$$

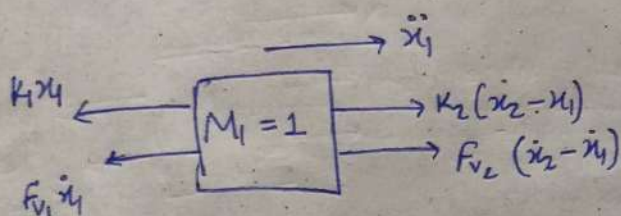


$$F - K_2(x_2 - x_1) - sF_{v2}(x_2 - x_1) - sF_{v3}x_2 = M_2 s^2 x_2$$

$$F - x_2(K_2 + sF_{v2} + sF_{v3}) - x_1(-K_2 - sF_{v2}) = M_2 s^2 x_2$$

$$F - x_2(5 + 3s + 2s) - x_1(-5 - 3s) = 2s^2 x_2$$

$$F = x_2(2s^2 + 5s + 5) + x_1(3s + 5) \quad \text{--- (1)}$$



$$K_2(x_2 - x_1) + F_{v2}s(x_2 - x_1) - K_1x_1 - F_{v1}x_1s = M_1s^2x_1$$

$$x_2(K_2 + sF_{v2}) = x_1(M_1s^2 + F_{v1}s + K_1 + F_{v2}s + K_2)$$

$$x_2(3s + 5) = x_1(s^2 + 6s + 9)$$

$$\Rightarrow x_2 = x_1 \left(\frac{s^2 + 6s + 9}{3s + 5} \right) \quad \text{--- (2)}$$

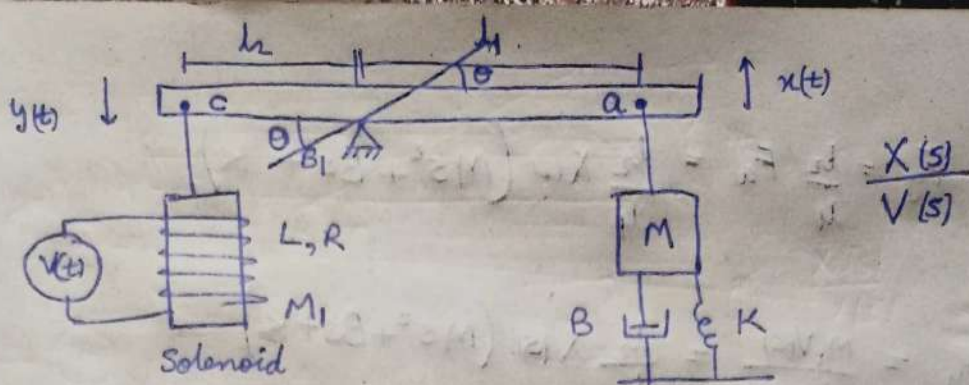
① and ②

$$F = x_1 \left(\frac{s^2 + 6s + 9}{3s + 5} \right) (2s^2 + 5s + 5) + x_1(3s + 5)$$

$$\frac{F}{x_1} = \frac{2s^4 + 17s^3 + 53s^2 + 75s + 45 - 9s^2 - 30s - 25}{3s + 5}$$

$$\frac{x_1}{F} = \frac{3s + 5}{2s^4 + 17s^3 + 44s^2 + 45s + 20}$$

⑧



Current & Voltage

$$V(t) - L \frac{di}{dt} - iR = 0$$

$$V(t) = L \frac{di}{dt} + iR$$

$$V(s) = sL I(s) + R I(s)$$

$$I(s) = \frac{V(s)}{sL + R}$$

Magnetic Force

$$F_y(t) = M_1 i(t)$$

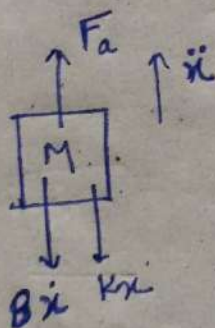
$$F_y(s) = M_1 I(s) = \frac{M_1 V(s)}{sL + R}$$

Lever Damping

$$F_y(t) = B_1 \frac{dy}{dt} \Rightarrow F_y(s) = B_1 Y(s) s$$

same = same

$$\frac{x(t)}{l_2} = \frac{y(t)}{l_1} \quad \text{①} \Rightarrow X(s) = \frac{l_1}{l_2} Y(s)$$



$$F_a(t) - B\dot{x}(t) - Kx(t) = M\ddot{x}(t)$$

$$\Rightarrow F_a(s) = s^2 M X + Bs X + KX$$

$$F_a(s) = X(s) (s^2 M + Bs + K)$$

differentiating ① $\Rightarrow$

$$\frac{\ddot{x}(t)}{l_1} = \frac{\ddot{y}(t)}{l_2}$$

and same = same

$$\Rightarrow \frac{F_a(s)}{l_1} = \frac{F_y(s)}{l_2}$$

$$F_y = \frac{l_2}{l_1} F_a = \frac{l_2}{l_1} X(s) (Ms^2 + Bs + K)$$

$$F_y = \frac{M_1 V(s)}{Ls + R} = \frac{l_2}{l_1} X(s) (Ms^2 + Bs + K)$$

$$\Rightarrow \frac{X(s)}{V(s)} = \frac{M_1 l_1}{l_2 (Ls + R) (Ms^2 + Bs + K)}$$

